

Jennifer Yang

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EDUCATION

Carnegie Mellon University

Master of Science in Robotics — GPA: 3.93/4.00

Bachelor of Science in Mechanical Engineering — GPA: 3.92/4.00

Pittsburgh, PA

Aug. 2024

May 2023

EXPERIENCE

Research Assistant

Carnegie Mellon University Zoom Lab

May 2023 – Aug. 2024

Pittsburgh, PA

- Built an affordable, easy-to-manufacture, omnidirectional quadruped robot with 3-DOF linear Delta actuators.
- Developed PyBullet simulations to test walking gaits, enabling rapid qualitative gait assessments.
- Applied nonlinear trajectory optimization to generate 4 different gaits with 4+ variations, and 1 rotation gait.
- Characterized robot performance with 50+ experiments, using computer vision to analyze position errors.
- Authored a comprehensive master's thesis and defended it in a formal presentation.

Mechanical Engineer Intern

Gecko Robotics

May 2022 – May 2023

Pittsburgh, PA

- Designed, prototyped, and assembled an End-of-Line tester for the TOKA 4 Robot Center Body, streamlining build and verification processes by reducing bottlenecks and saving 200+ hours throughout the build cycle.
- Conducted thermal testing to assess system reliability and prevent overheating during extended operational cycles.
- Collaborated with end-users to refine testing unit design, incorporating feedback to optimize ergonomics and functionality through rapid prototyping techniques.

Resident Assistant

Carnegie Mellon University

Aug. 2021 – May 2023

Pittsburgh, PA

- Fostered community engagement among 40+ residents by organizing events, leading House Council initiatives, and mediating conflicts to maintain a positive living environment.

Research Experiences for Undergraduates Student Intern

Oregon State University Laboratory for Robotics and Applied Mechanics

Jun. 2021 – Aug. 2021

Corvallis, OR

- Refined the sidewinding gait of a soft, pneumatic robot snake by assessing 2 control schemes and 8 flow rates.
- Enhanced understanding of its capabilities by evaluating performance across 3 terrains with 3 robot materials.

LEADERSHIP

Society of Women Engineers | Carnegie Mellon University

Aug. 2019 – Sep. 2022

- Coordinated the 2020 – 2022 STEM Career Fairs, managing catering for 200+ people, recruiting companies to attend, and partnering with CMU to promote the event.

PROJECTS

Haptic Mouse | Carnegie Mellon University

Mar. 2024 – May 2024

- Worked with 4 peers to implement haptic feedback on a computer mouse using a rotary Delta actuator.
- Conducted user studies to refine haptic feedback for improving learning experiences for visually impaired users.

PosChair: A Posture Enhancing Chair | Carnegie Mellon University

Aug. 2022 – Dec. 2022

- Worked with 4 peers to design and prototype a chair with an integrated sensor system for posture detection, haptic feedback for posture correction, and a GUI for visual feedback.

Bellcranks and Dampers System Lead | Carnegie Mellon Racing

Aug. 2021 – May 2022

- Designed and optimized the bellcranks and dampers system, ensuring it met specified design goals through kinematic simulations and stress analysis (FEA and hand calculations).
- CNC-machined components with tight tolerances and assembled the system onto the racecar.
- Documented design process and lessons learned in Confluence and the competition Design Binder.

SKILLS

Programming: Python, MATLAB, C, C++, Julia, Arduino, PyBullet

Software: CAD (SolidWorks), CAM (HSMWorks, MasterCAM), FEA (SolidWorks FEA, ANSYS)

Machines: CNC Mill, Vertical Mill, Lathe, 3D Printer, Laser Cutter

Languages: Mandarin (Fluent speaker), French (Conversant)